

second largest . java

{ 12, 35, 1, 10, 34, 1 }

by first Approach

★ by sorting()

c = [12, 35, 1, 10, 34, 1]

Array.sort

c = [1, 1, 10, 12, 34, 35]

int i = c.length - 2

i >= 0

i--

i = 6

6 - 2 = 4

4 >= 0 (t)

c[4] = 34

c[i] != c[c.length - 1]

c[6-1] != c[5]

c[4] != c[5]

34 != 35 (true)

return 34;

Approach 2:

one pass

{12, 35, 1, 10, 34, 1}

$a.length < 2$

$6 < 2$ (false) not executed

long first = long MIN Value.

Second = long min value

12 > first ✓ true

35 > 12 ✓ true

1 > 12 ✗ false

10 > 12 ✗ false

34 > 12 ✓ true

1 > 12 ✗ false

12 > first ✓

35 > 12 ✓

1 < 35 ✗

10 < 35 ✗

34 < 35 (true) ✓

first = 35

second 34

finally it print second 34

if all are same element in array {10, 10, 10}

1st Approach

after sorting

arr = [10, 10, 10]

$10 \neq 10$ ✗ false.

it return null



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2nd approach

initially

first = min_value

second = min_value

x = 10

10 > min true ✓

first = 10

x = 10

10 > 10 ✗

next

10 < 10 ✗

again 10

nothing change

return

first = 10

second = MIN

return

null;

Negative elements

arr $\{-5, -2, -9\}$

1st approach

Sorting

$[-9, -5, -2]$
0 1 2

Largest = -2

start at $i = 1$

$c[1] = -5$

$c[2] = -2$

$-5 \neq -2$ ✓

second Largest = -5

2nd approach

one pass

First = minvalue

second = minvalue.

first = -5

second = -2

~~-9 < -5~~

$-2 > -5$ true

Second = -5

first = -2



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$$x = -9$$

$$-9 < -2 \quad \checkmark$$

$$-9 > -5 \quad \times$$

nothing change

$$\text{first} = -2$$

$$\text{second} = -5$$

$$\text{output} = -5$$

* arr { 7, 7, 3 }

1st approach

sorting

arr [3, 7, 7]
0 1 2

Largest = 7

$$c[1] = 7$$

$$c[2] = 7$$

$$c[0] = 3$$

$$c[2] = 7$$

$$7 \neq 7 \quad \times$$

false

$$3 \neq 7 \quad \checkmark$$

true

return

~~c[i]~~

$$c[0] = 3$$

second largest



OSU IS TECHNICAL CAMPUS



onepass

first = min_value

second = min_value

x = 7

first = 7

second = min_value

x = 7

7 > 7 X

7 < 7 X

x = 3

3 > 7 X

3 < 7 ✓

3 > min ✓

first = 7

second = 3

second = 3

~~if~~ { 42 } to short-

sort

only one element so no second largest

return null;

~~if~~ { } array.length = 0

no second largest

return null;



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